

# Metrics

## Features:

Input image: 160x160 image RGB Format JPEG FILE

## Parameters:

| Layer (type)                                               | Output Shape                           | Param # |
|------------------------------------------------------------|----------------------------------------|---------|
| input_image ( <a href="#">InputLayer</a> )                 | ( <a href="#">None</a> , 160, 160, 3)  | 0       |
| random_flip ( <a href="#">RandomFlip</a> )                 | ( <a href="#">None</a> , 160, 160, 3)  | 0       |
| random_rotation ( <a href="#">RandomRotation</a> )         | ( <a href="#">None</a> , 160, 160, 3)  | 0       |
| random_translation ( <a href="#">RandomTranslation</a> )   | ( <a href="#">None</a> , 160, 160, 3)  | 0       |
| rescaling ( <a href="#">Rescaling</a> )                    | ( <a href="#">None</a> , 160, 160, 3)  | 0       |
| conv2d ( <a href="#">Conv2D</a> )                          | ( <a href="#">None</a> , 160, 160, 32) | 896     |
| batch_normalization ( <a href="#">BatchNormalization</a> ) | ( <a href="#">None</a> , 160, 160, 32) | 128     |
| conv2d_1 ( <a href="#">Conv2D</a> )                        | ( <a href="#">None</a> , 160, 160, 32) | 9,248   |
| max_pooling2d ( <a href="#">MaxPooling2D</a> )             | ( <a href="#">None</a> , 80, 80, 32)   | 0       |
| conv2d_2 ( <a href="#">Conv2D</a> )                        | ( <a href="#">None</a> , 80, 80, 128)  | 36,992  |

|                                                      |                     |        |
|------------------------------------------------------|---------------------|--------|
| batch_normalization_1<br>(BatchNormalization)        | (None, 80, 80, 128) | 512    |
| global_average_pooling2d<br>(GlobalAveragePooling2D) | (None, 128)         | 0      |
| dense (Dense)                                        | (None, 128)         | 16,512 |
| dropout (Dropout)                                    | (None, 128)         | 0      |
| predictions (Dense)                                  | (None, 1)           | 129    |

Model: "sequential\_1"

| Layer (type)        | Output Shape  | Param #    |
|---------------------|---------------|------------|
| flatten_1 (Flatten) | (None, 76800) | 0          |
| dense_4 (Dense)     | (None, 160)   | 12,288,160 |
| dense_5 (Dense)     | (None, 10)    | 1,610      |

Total params: 12,289,770 (46.88 MB)

Trainable params: 12,289,770 (46.88 MB)

Non-trainable params: 0 (0.00 B)

Output:

Is Wearing Mask True or False 1 = True 0 = False

## Hyperparameters

```
=====
TOP PERFORMING CONFIGURATION
=====
Optimal Learning Rate : 0.000143
Optimal Hidden Layers : 1
-----
=====
```

```
Trial 00 summary
Hyperparameters:
num_layers: 1
units_0: 160
activation_0: tanh
dropout_0: False
learning_rate: 0.0001428379365495633
Score: 0.5874999761581421
```